MODULE 5 FEMALE REPRODUCTIVE SYSTEM

The Female Reproductive System.

THE REPRODUCTIVE SYSTEM, FEMALE

Covers the ovaries, the uterine tubes and the path of the ovum, the uterus, the vagina and external genitalia, and the mammary glands. The focus is on the structures and the job each one does.

BY THE END

1. Identify the structures of the ovary and the uterine tubes.
2. Name the regions, ligaments, and wall layers of the uterus.
3. Identify the external genitalia and the structures of the mammary gland.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The female reproductive system

drsrennie-stack.github.io/new-build-bio4-solano/reproductive-female-concept-videos.html

The Female Reproductive System, an Overview

The female reproductive system produces oocytes, receives sperm, and supports a developing fetus. Its organs sort into the gonads, the genital tract, the external genitalia, and the mammary glands.

THE ORGANS OF THE SYSTEM

Structure	What it is
Ovaries	The female gonads; they produce oocytes and secrete hormones
Uterine tubes	The fallopian tubes; they carry the oocyte toward the uterus
Uterus	The muscular organ where a fertilized ovum implants and a fetus develops
Vagina	The canal that receives the penis and serves as the birth canal
Vulva	The external genitalia, collectively called the vulva or pudendum
Mammary glands	The milk-producing glands of the breasts

The Ovaries

The ovaries are paired, almond-shaped organs held beside the uterus by ligaments. Inside, the developing oocytes sit in structures called follicles.

THE OVARY, ITS ATTACHMENTS, AND ITS INTERIOR

Structure	What it is
Ovaries	The paired, almond-shaped female gonads, one on each side of the uterus
Broad ligament	The fold of peritoneum that drapes over the uterus and tubes; it attaches to the ovary by the mesovarium
Ovarian ligament	The cord that anchors the ovary to the uterus

Structure	What it is
Suspensory ligament	The fold that attaches the ovary to the pelvic wall
Germinal epithelium	The surface layer of low cuboidal epithelium covering the ovary
Tunica albuginea	The whitish capsule of dense connective tissue just beneath the germinal epithelium
Ovarian cortex	The region beneath the tunica albuginea that contains the ovarian follicles
Ovarian medulla	The deeper region of looser connective tissue holding blood vessels and nerves
Ovarian follicles	The structures in the cortex, each holding an oocyte at a stage of development
Mature follicle	The large, fluid-filled follicle ready to release its oocyte at ovulation
Corpus luteum	The yellow body; the structure left after ovulation, which later degenerates into a scar called the corpus albicans

KEY POINT

The ovary is read from the outside in: germinal epithelium, tunica albuginea, cortex with the follicles, then medulla with the vessels. The follicles you are asked to find are always in the cortex.

The Uterine Tubes and the Path of the Ovum

The uterine tubes catch the released oocyte and carry it toward the uterus. Fertilization, when it happens, usually happens here.

THE REGIONS OF THE UTERINE TUBE

Structure	What it is
Infundibulum	The funnel-shaped end of the tube, open to the pelvic cavity near the ovary
Fimbriae	The fringe of finger-like projections at the infundibulum that sweep the oocyte into the tube
Ampulla	The widest and longest portion of the tube, where fertilization usually occurs
Isthmus	The narrow, thick-walled portion that joins the tube to the uterus
Lining of the tube	Ciliated and secretory cells whose cilia and fluid move the oocyte toward the uterus

The ovum follows one route from the ovary to the uterus.

1. **Ovary** the mature follicle releases the secondary oocyte at ovulation
2. **Fimbriae and infundibulum** the fimbriae sweep the oocyte into the funnel of the tube
3. **Ampulla** the oocyte travels through the widest part of the tube, where sperm may fertilize it
4. **Isthmus** the oocyte or early embryo passes through the narrow part of the tube
5. **Uterus** the oocyte or embryo reaches the uterine cavity

The Uterus

The uterus is a thick-walled, pear-shaped organ between the bladder and the rectum. Ligaments hold it in place and three layers build its wall.

THE REGIONS OF THE UTERUS AND THE LIGAMENTS THAT HOLD IT

Structure	What it is
Fundus	The dome-shaped superior portion of the uterus, above the entry of the tubes
Body	The tapered central portion of the uterus
Cervix	The narrow inferior portion that opens into the vagina
Uterine cavity	The interior space of the body of the uterus
Cervical canal	The interior of the cervix, opening to the uterine cavity at the internal os and to the vagina at the external os
Broad ligaments	The paired folds of peritoneum attaching the uterus to the pelvic walls
Uterosacral ligaments	The paired folds connecting the uterus to the sacrum
Cardinal ligaments	The paired ligaments extending from the pelvic wall to the cervix and vagina
Round ligaments	The paired fibrous cords running from the uterus to the labia majora

The three layers of the uterine wall

The wall of the uterus has three layers. Compare them.

THE LAYERS OF THE UTERINE WALL

Layer	What it is
Perimetrium	The outer serous layer, part of the visceral peritoneum
Myometrium	The thick middle layer of smooth muscle that contracts during childbirth
Endometrium	The inner vascular lining; its stratum functionalis sheds in menstruation and its stratum basalis rebuilds it

The Vagina and External Genitalia

The vagina connects the cervix to the outside. The external genitalia, the vulva, surround its opening.

THE VAGINA AND THE STRUCTURES OF THE VULVA

Structure	What it is
Vagina	The fibromuscular canal from the cervix to the exterior; the birth canal
Fornix	The recess of the vagina that surrounds the cervix
Hymen	The thin fold of mucous membrane that partly closes the vaginal opening
Mons pubis	The cushion of fatty tissue over the pubic symphysis
Labia majora	The two larger, hair-covered folds of skin; homologous to the scrotum
Labia minora	The two smaller, hairless folds within the labia majora
Clitoris	The small mass of erectile tissue at the anterior junction of the labia minora; homologous to the penis
Vestibule	The region between the labia minora, holding the vaginal and urethral openings
Greater vestibular glands	The Bartholin's glands beside the vaginal opening, which secrete lubricating mucus
Perineum	The diamond-shaped region holding the external genitalia and anus, split into a urogenital triangle and an anal triangle

The Mammary Glands

The mammary glands lie in the breasts, over the chest muscles. They are modified sweat glands that produce milk.

THE STRUCTURES OF THE MAMMARY GLAND

Structure	What it is
Mammary glands	The modified sweat glands of the breast that produce milk
Lobes	The 15 to 20 or more compartments of the mammary gland, each divided into lobules
Lobules	The subdivisions of a lobe, containing the milk-secreting alveoli
Alveoli	The small sacs of glandular tissue that secrete milk
Lactiferous ducts	The ducts that carry milk from the lobes toward the nipple
Nipple	The pigmented projection where the lactiferous ducts open
Areola	The circular pigmented area of skin around the nipple
Suspensory ligaments	The Cooper's ligaments that support the breast

KEY POINT

The mammary gland nests in one direction: alveoli inside lobules, lobules inside lobes, lobes draining by lactiferous ducts to the nipple. Name them in that order and you have named the path milk takes.

Common Disorders of the Female Reproductive System

Compare the common disorders of the female reproductive system by the structure each one affects.

THE COMMON DISORDERS

Disorder	What it is
Endometriosis	Endometrial tissue grows outside the uterus, on the ovaries or pelvic lining
Ovarian cysts	Fluid-filled sacs that form on or in an ovary
Uterine fibroids	Non-cancerous tumors of the smooth muscle of the myometrium
Pelvic inflammatory disease	An infection that spreads through the uterus, tubes, and ovaries
Cervical cancer	A malignant tumor of the cervix
Ectopic pregnancy	An embryo implants outside the uterus, most often in a uterine tube
Breast cancer	A malignant tumor of the mammary gland tissue

OPEN PATHWAYS AND SHED LININGS

The female tract has an open route to the abdomen. The **infundibulum** opens into the pelvic cavity rather than sealing onto the ovary, which is how an oocyte is caught, but also how infection can climb from the vagina through the uterus and tubes to cause **pelvic inflammatory disease**.

The same open route explains **ectopic pregnancy**. If a fertilized ovum is delayed in the uterine tube, it can implant there instead of in the uterus, in a narrow tube never built to hold it.

The **endometrium** is built in two layers for a reason. The stratum functionalis is shed and rebuilt each cycle from the permanent stratum basalis, so the lining can renew itself month after month.